

- 2 A scientist is investigating the size of shells at various beach locations. She selects four beach locations and takes a random sample of shells from each of these beaches. She classifies each shell as large or small. Her results are summarised in the following table.

		Beach location				Total
		<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	
Size of shell	Large	68	69	96	81	314
	Small	28	55	64	39	186
	Total	96	124	160	120	500

Test, at the 10% significance level, whether the size of shell is independent of the beach location. [7]

[illegible]